

## Compact Closure: Snake Equation, Valency, and Four Complexity Metrics

# Compact Closure: Snake Equation, Valency, and Four Complexity Metrics I

**Where we are in the argument.** ??–?? made each sentence a string diagram that simultaneously expresses syntax and semantics. This chapter zooms in on the *axiom* that makes the machine run — the compact-closure snake equation  $(\varepsilon_n \otimes 1_n) \circ (1_n \otimes \eta_n) = 1_n$  — and uses its consequences to define four graded complexity metrics (word count, cup count, cap count, depth) that turn “how hard is this sentence to parse?” from a qualitative question into a computable one, feeding directly into the enriched hom-values of ??.

# The Snake Equation Powers Every Pregroup Contraction I

# The Snake Equation Powers Every Pregroup Contraction II

## The Snake Equation (Zigzag Identity)

The essential algebraic engine driving DisCoCat is the strict **compact closure** of the governing pregroup category. For every lexical type  $n$ , the corresponding adjunction maps— $\eta_n : 1 \rightarrow n \otimes n^r$  (the cap expansion) and  $\varepsilon_n : n^r \otimes n \rightarrow 1$  (the cup contraction)—must mathematically satisfy the fundamental **snake equation** (also known as the topological zigzag identity):

$$(\varepsilon_n \otimes 1_n) \circ (1_n \otimes \eta_n) = 1_n \quad (1)$$

Translated into intuitive string-diagrammatic terms, a cup geometrically composed with a cap bridging adjacent wires immediately “straightens out” into a solid identity wire—a topological zigzag seamlessly canceling into a continuous straight line. This foundational axiom is not merely a geometric curiosity: it physically functions as the *engine* powering every valid pregroup type reduction. Every recorded grammatical contraction (e.g., a noun canceling against an active verb argument slot) constitutes a